

AP Statistics FRQ Practice Exam 08

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AP Statistics FRQ Practice Exam 8

APSTAT-FRQ8-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 3, Inference for Categorical Data: Proportions

FRQ Type: Formulate Questions and Collect Data

Prompt:

A city youth office wants to test whether text-message reminders increase attendance at orientation sessions for families registered at after-school centers. The office randomly selects 24 of its 80 centers. It randomly assigns 12 selected centers to send a text reminder 24 hours before orientation and the other 12 centers to continue using the standard email notice. Every registered family at a center receives that center's assigned notice.

Treatment assigned to center	Number of centers	Mean center attendance rate
Text reminder	12	68%
Standard email	12	59%

- (a) Identify the population, sample, and experimental units. Explain why individual families are not the experimental units.
- (b) Identify the treatments and response variable. State a statistical question the experiment is designed to answer.
- (c) Describe a valid random-assignment procedure for the 24 centers. Explain the purpose of random assignment.
- (d) Calculate and interpret the observed treatment difference. Explain why an analysis should account for centers as the units of assignment rather than treat every family response as an independent treatment assignment.
- (e) State the causal and generalization conclusions the design could support if the attendance-rate difference is judged meaningful. Identify one factor that should be kept consistent across treatment groups.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies the population as all 80 city after-school centers and the sample as the 24 randomly selected centers.
- Point 2: Identifies centers as the experimental units because the notice type is assigned to whole centers; families receive, but are not individually assigned to, the treatment.

(b) 2 points

- Point 1: Identifies text reminder and standard email as the treatments and each center's family attendance rate as the quantitative response.
- Point 2: States an appropriate comparative question about whether text reminders change the mean center attendance rate compared with standard email.

(c) 2 points

- Point 1: Describes a valid chance process, such as randomly shuffling the 24 center labels and assigning the first 12 to text and the remaining 12 to email.
- Point 2: Explains that random assignment tends to balance center characteristics across groups and supports a cause-and-effect comparison of notice types.

(d) 2 points

- Point 1: Calculates $68\% - 59\% = 9$ percentage points and interprets it as the text-assigned centers having a mean attendance rate 9 percentage points higher than the email-assigned centers.
- Point 2: Explains that families within a center share the same assignment and may have correlated outcomes, so treating all family responses as independently randomized would exaggerate the amount of independent information.

(e) 2 points

- Point 1: States that random assignment supports a causal conclusion about notice type for centers like those studied, and random selection supports generalization to the 80 city centers.
- Point 2: Identifies a relevant control, such as keeping the orientation content, timing, registration rules, and follow-up procedures the same across groups.

Model Guidance:

The treatment was assigned at the center level, so the 24 centers are the experimental units. The family-level attendance outcomes are nested within those units. Randomly divide the selected centers into two equal groups before the notices are sent, and hold other orientation procedures constant.

The observed difference is 9 percentage points in favor of text reminders. An analysis that pretends each family was separately randomized would ignore the similarity of families exposed to the same center, staff, schedule, and notice. Because centers were randomly selected and then randomly assigned, a meaningful difference can be attributed to notice type and generalized to the city's 80 centers, subject to consistent implementation.

Expected Points:

10

APSTAT-FRQ8-002

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: None

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

A mountain resort records the number of minutes needed to reopen a ski lift after each of 15 sensor alerts during one winter month. The recorded times, in increasing order, are

3, 4, 4, 5, 5, 6, 6, 7, 8, 9, 10, 12, 15, 28, 41.

(a) On paper, construct a histogram using intervals $0 < 10$, $10 < 20$, $20 < 30$, $30 < 40$, and $40 < 50$ minutes. Label both axes and give the frequency in each interval.

(b) Describe the distribution's shape, center, spread, and unusual features.

(c) Calculate the mean and median. Explain why the two measures differ and identify which is the more representative center for these data.

(d) Calculate Q_1 , Q_3 , the IQR, and the $1.5 \times \text{IQR}$ outlier fences. Identify any potential outliers.

(e) An audit determines that the value 41 was a data-entry error and should have been 16. Calculate the corrected mean and median. Explain what the changes demonstrate about resistance.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Draws a histogram with reopening time on the horizontal axis, frequency on the vertical axis, and the five specified equal-width intervals.
- Point 2: Gives frequencies 10, 3, 1, 0, and 1 for the intervals $0 < 10$ through $40 < 50$, respectively.

(b) 2 points

- Point 1: Describes the distribution as strongly right-skewed, with most times below 10 minutes and a long upper tail.
- Point 2: Places the center near the median of 7 minutes, describes the spread as extending from 3 to 41 minutes, and notes 28 and 41 as unusual relative to the main cluster.

(c) 2 points

- Point 1: Calculates $\bar{x} = 163/15 \approx 10.87$ minutes and the median as 7 minutes.
- Point 2: Explains that the large upper values pull the mean above the median and selects the median as a more representative center for the strongly right-skewed data.

(d) 2 points

- Point 1: Calculates $Q_1 = 5$, $Q_3 = 12$, $\text{IQR} = 7$, and fences $5 - 1.5(7) = -5.5$ and $12 + 1.5(7) = 22.5$ minutes.
- Point 2: Identifies 28 and 41 minutes as potential high outliers because both exceed 22.5.

(e) 2 points

- Point 1: Calculates the corrected mean as $(163 - 41 + 16)/15 = 138/15 = 9.2$ minutes and the corrected median as 7 minutes.

· Point 2: Explains that the mean changes substantially while the median does not, demonstrating that the median is resistant to an extreme-value correction whereas the mean is not.

Model Guidance:

The histogram has a dominant first bar of height 10, followed by heights 3, 1, 0, and 1. That display, together with the ordered list, shows a long right tail. The mean is about 10.87 minutes, well above the median of 7 minutes because 28 and 41 pull it upward.

Using the median-of-halves convention gives $Q1 = 5$ and $Q3 = 12$, so the upper outlier fence is 22.5 minutes. Both 28 and 41 are flagged. Replacing 41 with 16 lowers the mean to 9.2 but leaves the median at 7. The comparison illustrates why the median and IQR are preferred when describing center and spread for data with strong skew or extreme values.

Expected Points:

10

APSTAT-FRQ8-003

Section: FRQ

Type: frq

Primary Unit: Unit 4, Inference for Quantitative Data: Means

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

Ornithologists compare the mean hovering duration per feeding visit for two hummingbird species. They take independent simple random samples, with one observation from each selected adult bird. Each species population is much larger than 10 times its sample size. Side-by-side dotplots are roughly symmetric with no apparent outliers.

Species	n	Sample mean, $\bar{x}$	Sample standard deviation, s
Coppercrest	20	31.6 seconds	2.4 seconds
Violetcap	18	29.9 seconds	2.1 seconds

At the $\alpha = 0.05$ significance level, test whether the true mean hovering duration is greater for Coppercrest hummingbirds than for Violetcap hummingbirds. Use $df = 35$ for the two-sample t procedure.

- Define the population parameters and state the hypotheses.
- Identify the appropriate inference procedure and check its conditions.
- Calculate the standard error, test statistic, and p-value.
- State a conclusion in context using the $\alpha = 0.05$ significance level.

(e) Explain why the conclusion does not establish that species membership itself causes a difference in hovering duration.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Defines μ_C and μ_V as the true mean hovering durations per feeding visit for adult Coppercrest and Violetcap hummingbirds, respectively.
- Point 2: States $H_0: \mu_C - \mu_V = 0$ and $H_a: \mu_C - \mu_V > 0$.

(b) 3 points

- Point 1: Selects an independent two-sample t test for a difference in population means.
- Point 2: Cites the independent SRSs and verifies the 10% condition for each species.
- Point 3: Uses the roughly symmetric, outlier-free dotplots to justify the t procedure for the two modest sample sizes.

(c) 3 points

- Point 1: Calculates $SE = \sqrt{[(2.4^2/20) + (2.1^2/18)]} \approx 0.7301$ second.
- Point 2: Calculates $t = [(31.6 - 29.9) - 0]/0.7301 \approx 2.33$ with $df = 35$.
- Point 3: Calculates the right-tail p -value $P(T_{35} \geq 2.33) \approx 0.0129$.

(d) 1 point

- Point 1: Because $0.0129 < 0.05$, rejects H_0 and concludes that the samples provide convincing evidence that adult Coppercrest hummingbirds have a greater true mean hovering duration per feeding visit than adult Violetcap hummingbirds.

(e) 1 point

- Point 1: Explains that species was observed rather than randomly assigned and may be associated with body size, habitat, flower type, or other variables; the study supports a population comparison but not a manipulable causal effect of species.

Model Guidance:

The response is quantitative and the two samples contain different birds, so use an independent two-sample t test. Random sampling, small sampling fractions, and the roughly symmetric distributions without outliers support the procedure.

The observed mean difference is 1.7 seconds. Dividing by the standard error of about 0.7301 gives $t \approx 2.33$ with a one-sided p -value near 0.0129. This is below 0.05, supporting a difference in the stated direction. Random sampling permits inference to the two adult-bird populations, but an observational comparison of naturally occurring species cannot assign causation to species membership or rule out associated biological and environmental variables.

Expected Points:

10

APSTAT-FRQ8-004

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A food-truck cooperative must choose one inventory plan for an upcoming festival day. Based on many comparable festival days, demand has the following distribution. The table gives net profit under each plan.

Demand state	Probability	Standard plan profit	Expanded plan profit
Low	0.30	\$300	-\$400
Typical	0.45	\$900	\$1,200
High	0.25	\$1,300	\$2,500

(a) Verify that the probabilities form a valid distribution. Calculate the expected net profit for each inventory plan.

(b) For each plan, calculate the probability of earning at least \$1,000. Also identify the probability of a net loss. Compare the plans using these two risk measures.

(c) On the morning of the festival, a reliable forecast of heavy rain changes the conditional demand probabilities to 0.60 for Low, 0.30 for Typical, and 0.10 for High. Recalculate the expected net profit for each plan under the rain forecast.

(d) A manager says, "The Expanded plan is always the better choice because its original expected profit is higher." Evaluate the statement using the unconditional distribution, the rain-forecast distribution, and the possibility of a loss.

(e) If the cooperative must choose before receiving any weather forecast and its sole objective is to maximize long-run mean profit over many comparable festival days, identify the preferred plan. Explain why that recommendation is not a guarantee about profit on this one festival day and how a different attitude toward risk could change the decision.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Verifies that $0.30 + 0.45 + 0.25 = 1.00$ and that all probabilities lie from 0 to 1.
- Point 2: Calculates $E(\text{Standard}) = 0.30(300) + 0.45(900) + 0.25(1,300) = \820 and $E(\text{Expanded}) = 0.30(-400) + 0.45(1,200) + 0.25(2,500) = \$1,045$.

(b) 2 points

- Point 1: Calculates $P(\text{profit} \geq \$1,000) = 0.25$ for Standard and $0.45 + 0.25 = 0.70$ for Expanded.
- Point 2: Identifies loss probabilities 0 for Standard and 0.30 for Expanded, then explains that Expanded has a larger chance of a high profit but also introduces a 30% loss risk.

(c) 2 points

- Point 1: Calculates $E(\text{Standard} \mid \text{heavy-rain forecast}) = 0.60(300) + 0.30(900) + 0.10(1,300) = \580 .

· Point 2: Calculates $E(\text{Expanded} \mid \text{heavy-rain forecast}) = 0.60(-400) + 0.30(1,200) + 0.10(2,500) = \370 .

(d) 2 points

· Point 1: Notes that Expanded has the larger unconditional expected profit, \$1,045 versus \$820, but Standard has the larger conditional expected profit under the heavy-rain forecast, \$580 versus \$370.

· Point 2: Rejects “always better” because the preferred expected-value choice changes with relevant conditional information and because Expanded carries a loss probability that Standard does not.

(e) 2 points

· Point 1: Selects Expanded when choosing before a forecast under the stated long-run mean-profit objective because $\$1,045 > \820 .

· Point 2: Explains that expected value is a long-run average rather than a guaranteed single-day result and that a cooperative strongly concerned about avoiding losses could reasonably prefer Standard despite its lower expectation.

Model Guidance:

The original demand probabilities sum to 1. Expected profit weights each possible profit by its demand probability. Expanded has the larger original expectation, \$1,045 versus \$820, and a 70% chance to earn at least \$1,000, but it also has a 30% chance of losing \$400. Standard never loses money in the table.

Conditional information can change a rational decision. Under the heavy-rain probabilities, Standard's expected profit is \$580 and Expanded's is only \$370. Thus, Expanded is the expected-profit choice before any forecast, but it is not universally superior. Expected value describes average performance across repeated comparable days; the cooperative's tolerance for loss and any reliable forecast also matter for a one-day decision.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ8

· APSTAT-FRQ8-001: Unit 1 primary; Unit 3 secondary; cluster-randomized experiment on 24 centers randomly selected from 80 city after-school centers, assigning 12 centers to text reminders and 12 to standard email, with mean center family-attendance rates 68% and 59%; identify centers rather than families as experimental units, formulate the attendance question, randomize whole centers, calculate a 9-percentage-point difference, explain within-center dependence, and separate causal from generalization scope; targets pseudoreplication from treating nested family outcomes as independently assigned.

· APSTAT-FRQ8-002: Unit 1 primary; no secondary unit; ordered one-variable dataset of 15 ski-lift reopening times 3, 4, 4, 5, 5, 6, 6, 7, 8, 9, 10, 12, 15, 28, and 41 minutes; construct a five-bin histogram with frequencies 10, 3, 1, 0, 1, describe strong right skew, calculate mean 10.87 and median 7, calculate $Q1 = 5$, $Q3 = 12$, $IQR = 7$ and fences $-5.5/22.5$, flag 28 and 41, then replace 41 with 16 to contrast mean 9.2 with unchanged median; targets relying on the mean for skewed data and confusing resistance with no change in every statistic.

· APSTAT-FRQ8-003: Unit 4 primary; Unit 1 secondary; one-sided independent two-sample t test comparing adult Coppercrest and Violetcap hummingbird mean hovering durations, using independent SRS summaries $n = 20$, $\bar{x} = 31.6$, $s = 2.4$ and $n = 18$, $\bar{x} = 29.9$, $s = 2.1$; check SRS/independence/10%/shape, use $df = 35$, calculate $SE \approx 0.7301$, $t \approx 2.33$ and $p \approx 0.0129$, reject at α

= 0.05, and distinguish population comparison from causation; targets pooling standard deviations without instruction and treating species comparison as randomized treatment.

· APSTAT-FRQ8-004: Unit 2 primary; Unit 1 secondary; food-truck inventory decision with Low/Typical/High demand probabilities 0.30/0.45/0.25 and Standard versus Expanded profit table; verify the distribution, calculate expected profits \$820 and \$1,045, compare target-profit and loss probabilities, update expectations to \$580 and \$370 under heavy-rain conditional probabilities 0.60/0.30/0.10, and evaluate “always better” and long-run expected-value claims under risk; targets treating expected value as a single-event guarantee and ignoring conditional information or downside probability.